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AI-generated content and academic norms: perspectives and reflections of college students

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ABSTRACT

Generative AI tools, such as ChatGPT, are transforming education and sparking debates surrounding academic integrity. We surveyed college students from diverse cultural backgrounds to explore the ethical implications of AI in academic settings. This analysis included a review of international cases and relevant literature. The survey revealed a range of student perspectives: some view AI-generated content as fair use, while others express concerns that it may encourage plagiarism and undermine critical thinking. Students also shared varied opinions on the regulation of AI use and the importance of fostering awareness of academic integrity. The emergence of ChatGPT has also presented challenges for educators in their teaching efforts. Our study provides data to enhance the understanding of plagiarism and assist in establishing academic norms. Based on the survey results, we propose measures to promote regulatory policies and teaching adjustments, as well as to develop standardized guidelines for AI use. Our goal is to establish comprehensive norms for the responsible and effective application of AI in higher education.

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Introduction

The era of artificial intelligence (AI) is developing rapidly. From generative artificial intelligence represented by ChatGPT, which is expected to have a significant impact on all digital advertising fields globally (Baek, 2023), to the fact that generative artificial intelligence methods can create designs such as small molecule drugs and proteins by analyzing different data patterns (including images and sequences) (Wang et al., 2023). From the extraordinary ability of AlphaGo in the game of Go to the increasingly complex autonomous vehicles, intelligent AI applications are subtly changing the way we learn, work and live.

Currently, the application of artificial intelligence in education is an ever-changing target. The best practices for its utilization will be as diverse and subtle as the disciplines to which it is applied (Sarah Wells, 2024). The research findings on the potential application of artificial intelligence in education to enhance educational quality and boost students' knowledge levels indicate that the knowledge level of students receiving personalized AI-enhanced learning has been significantly improved (Liu & Yushchik, 2024). However, for scholars, their work may become more arduous due to artificial intelligence. As it will make more articles appear more refined, with more appropriate citations and more beautiful illustrations. Inevitably, there will be some “fake news” and “alternative facts” intermingled among real scientific breakthroughs (Wong et al., 2024).

Therefore, the rapid development of AI has made it useful in education; however, it also gives rise to concerns regarding academic integrity. For example, what impact will AI generation technology have on current education? How do students and educators address these effects? What strategies should we take to effectively integrate AI with students and educators? These questions are all worth exploring and thinking about.

This paper examines various issues related to AI and higher education in China from the perspective of students, utilizing data collected through an online survey. The aim is to understand students' views and to analyze and summarize the ethical challenges posed by the use of AI tools in university education. The impact of AI on higher education in China is explored, particularly in relation to academic integrity. Additionally, the paper offers suggestions for educators on how to uphold the fundamental principles of academic integrity. Given that attitudes towards AI's role in academic creation may differ across national cultures, this study also reviews the norms of various countries regarding AI's involvement in academic work, thereby exploring the perceptions of different nations in this context.

This study is dedicated to exploring the interconnected relationship between integrity issues, such as academic plagiarism, and artificial intelligence (AI) technology. It emphasizes the importance of adapting to contemporary advancements and aims to establish a robust higher education system that incorporates AI.

Methods

This study examines the perspectives of Chinese university students on the impact of ChatGPT on academic integrity through a questionnaire survey. University students serve as our research subjects and are also the primary users of artificial intelligence. By employing a questionnaire survey, we can ensure that our data and analysis results are more reliable.

The study primarily focuses on undergraduate students, examining the views of different grades and majors within a university regarding the role of artificial intelligence in academic creation. We surveyed 162 university students, including 149 undergraduates and 13 graduate students. All participants were at least 18 years old and provided consent for the purpose and design of this research. Prior to data collection, we obtained informed consent from all participants. All research data will be used solely for academic purposes and will remain anonymous.

Taking into account the cultural differences between countries, we conducted a literature review to analyze case studies on the role of artificial intelligence in academic creation in various countries, aiming to collect and summarize the views of college students from different nations on this topic.

Data collection and analysis

We collected data through a questionnaire survey. The questionnaire featured an innovative gamified design, incorporating visual elements inspired by local cultural landmarks to enhance engagement. Additionally, we implemented incentive mechanisms to boost user participation. Upon completing the questionnaire, respondents were presented with an achievement badge reward system: they automatically received a guaranteed badge ("AI Ethics Pioneer") based on their responses, along with an additional randomly selected badge from a pool of over 20 pre-designed titles (categorized into humorous, academic, and rare/mysterious types, with probabilities of 30%, 40%, and 30%, respectively). These badges could be shared on social media platforms.

After finalizing the questionnaire design, we distributed the survey links to students both online and in person during breaks between classes. Students could access the questionnaire via the link on either desktop or mobile devices. During the offline data collection process, students provided positive feedback, noting the user-friendly interface and ease of operation.

The correlation was tested using a two-tailed *t*-test in SPSS, while differences were analyzed using an independent samples *t*-test. To address the uncertainty in the questionnaire survey, Cronbach's

Table 1. Credibility analysis.

Subscale	N of items	Cronbach's α	Mean	SD
Basic understanding of AI	3	0.611	6.8	1.622
Opinions on AI regulation	3	0.389	6.1	1.674
Classification of AI cheating behaviours	5	0.671	11.51	2.552

Overall Cronbach's $\alpha = 0.602$.

alpha analysis was ultimately employed to assess reliability. The survey indicates that the overall Cronbach's alpha coefficient is 0.602, suggesting that the survey possesses a moderate level of reliability. It effectively assesses college students' perceptions of AI participation in academic creation, thereby providing a solid foundation for subsequent investigations (see Table 1).

Result

Variable analysis

The degree of exposure to artificial intelligence varies across different grade levels and academic majors. For example, students in lower grades have limited exposure to artificial intelligence, as they are primarily focused on foundational courses. In contrast, senior students may engage more extensively with artificial intelligence through club activities and thesis projects. Additionally, the level of exposure to artificial intelligence can differ significantly among various majors. Students majoring in fields such as artificial intelligence, automation, and computer science will receive more in-depth education related to artificial intelligence, while some students majoring in liberal arts may only have a basic understanding of the application of artificial intelligence. To investigate this relationship, we will analyze grade level and major as variables. (QN represents the topic of the questionnaire, with $N = 1, 2, 3, \dots$)

As shown in Table 2, engineering majors comprised the largest proportion of respondents in this survey, totalling 91 individuals. Among these, first-year engineering students represented 47.3% of the total number of students across all grades, making them the largest subgroup within the engineering discipline. In contrast, the medical field had the smallest representation, with only 5 students. Within this group, fourth-year medical students accounted for 60% of their grade level, representing the largest cohort in the medical major. In terms of grade distribution, first-year students had the highest number, totalling 76 individuals. Science majors constituted the highest proportion among postgraduate students, reaching 16.7%. Additional information can be described in a similar manner in the table (see Table 2).

In Table 3, Q1 represents the grade level, Q2 denotes the professional type, Q3 indicates the understanding of the application of artificial intelligence (AI) in content creation, and Q5 assesses the use of AI tools to assist with academic writing or research. The table reveals that the average values for each item fall within a relatively moderate range, reflecting the stability and rationality of the data. The moderate standard deviation suggests that the survey effectively captures the attitudes and variations among different students. Notably, the coefficients for Q1 and Q5, as well as the correlation coefficients for Q3 and Q5, were significantly correlated. This indicates that students'

Table 2. Variable distribution.

Grade	Specialized field					Total
	Liberal arts	Science	Engineering	Medicine	Other	
Freshman	18.2%	46.7%	47.3%	20%	64%	76
Sophomore	9.1%	13.3%	8.8%	8.8%	24.0%	20
Junior	27.3%	10.0%	8.8%	0.0%	4.0%	15
Senior	45.5%	13.3%	26.4%	60.0%	8.0%	38
Postgraduate student	0.0%	16.7%	8.8%	0.0%	0.0%	13
Total	11	30	91	5	25	162

Table 3. Correlation between grade-level specialization and basic AI awareness (Pearson coefficient).

	Mean	SD	Q1	Q2	Q3	Q5
Q1	2.33	1.457	1.000	-.210**	-.228*	-.514**
Q2	3.02	1.054		1.000	.040	.126
Q3	2.12	.685			1.000	.339**
Q5	2.17	.816				1.000

Note: **The correlation is significant at the 0.01 level (two-tailed).

grade levels and their cognitive understanding of AI influence their behaviour regarding the use of tools in academic contexts, thereby confirming the relationship between cognition and practice (see Table 3).

As shown in Table 4, Q3 indicates your understanding of the application of artificial intelligence (AI) in content creation. Q5 reflects whether you have attempted to use AI tools to assist with academic writing or research. Q12 expresses your opinion on a school's explicit prohibition of AI-generated content. The results found differences in the cognitive application and practice of AI content creation among students of varying grade levels; however, there was consistency in their attitudes towards academic norms. The data indicates that senior students (those in grades above junior) possess a significantly higher understanding of AI applications in content creation compared to junior students. This disparity may be attributed to their more extensive academic research experience. Additionally, senior students demonstrated greater initiative in utilizing AI tools to assist with academic writing and research, reflecting their practical need for intelligent tools in complex academic tasks. It is noteworthy that, despite the differences in application levels, both groups of students exhibited a statistically insignificant stance regarding the school's prohibitive attitude towards uncensored AI-generated content. This consensus may stem from shared values of academic integrity across all grade levels, suggesting that students generally agree on the necessity of reviewing and modifying AI-generated content in accordance with academic standards, regardless of their level of technological engagement.

Awareness and use of AI tools

The survey results show that 98.77% of students report having some knowledge of AI tools, such as ChatGPT, which can generate text and provide answers. Respondents self-rated their AI knowledge as follows: 16.67% "know very well," 55.56% "have a certain understanding," and 26.54% "have heard of it but don't know much about it." This reflects their status as digital natives, as they find it easy to master and utilize these tools (see Figure 1).

96.3% of students have utilized AI tools to get information or receive responses. Among them, only 3.7% of the people know nothing about this AI tool. The majority of participants reported extensive experience with search engines, emotional intelligence tools, and various AI applications, including ChatGPT. They are well-acquainted with and frequently use a range of tools, such as New Bing, Stable Diffusion, and Midjourney. Participants expressed that these tools facilitated the rapid generation of text, provided answers to questions, and assisted with text editing (see Figure 2).

Table 4. Difference analysis (*T*-test).

	Grade	<i>N</i>	Mean	SD	<i>t</i>	<i>p</i>
Q3	≥Junior	66	1.97	.679	-2.404	0.017*
	<Junior	96	2.23	.672		
Q5	≥Junior	66	1.71	.718	-6.742	<0.001***
	<Junior	96	2.49	.725		
Q12	≥Junior	66	2.41	.803	-1.051	.295
	<Junior	96	2.54	.767		

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

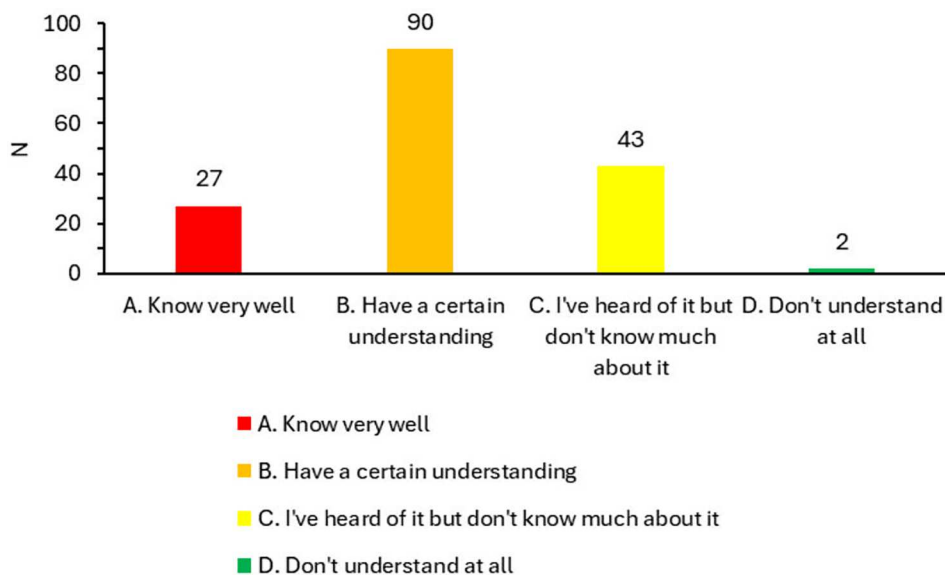


Figure 1. Statistics of AI understanding.

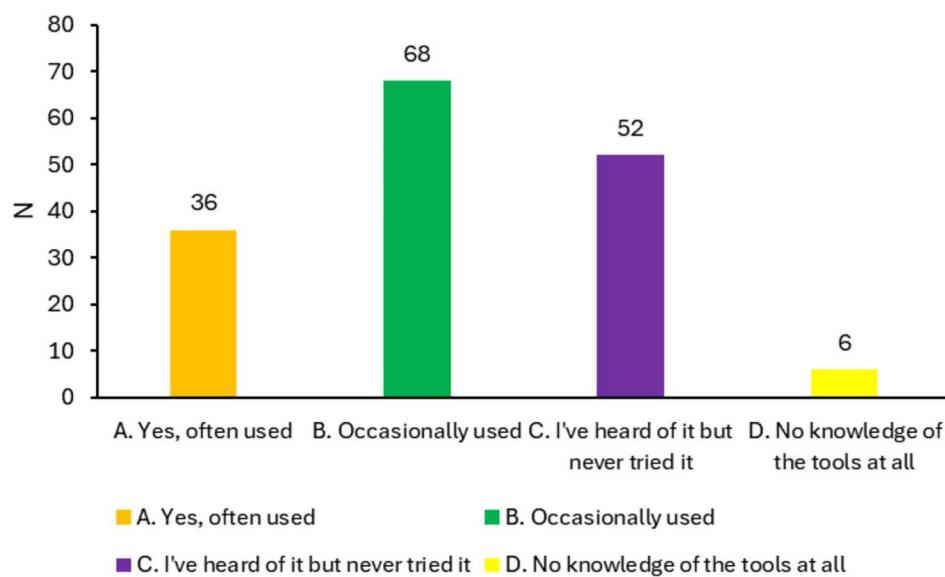


Figure 2. Statistics on the use of artificial intelligence.

When it comes to using tools like ChatGPT, the main purpose for most users is homework assistance and knowledge acquisition, while a smaller percentage utilizes the platform for entertainment and casual conversation.

Perceptions of academic integrity

According to the survey results presented in [Figures 3–6](#), there is a divergence of opinions among respondents regarding whether the use of AI-generated content should adhere to academic

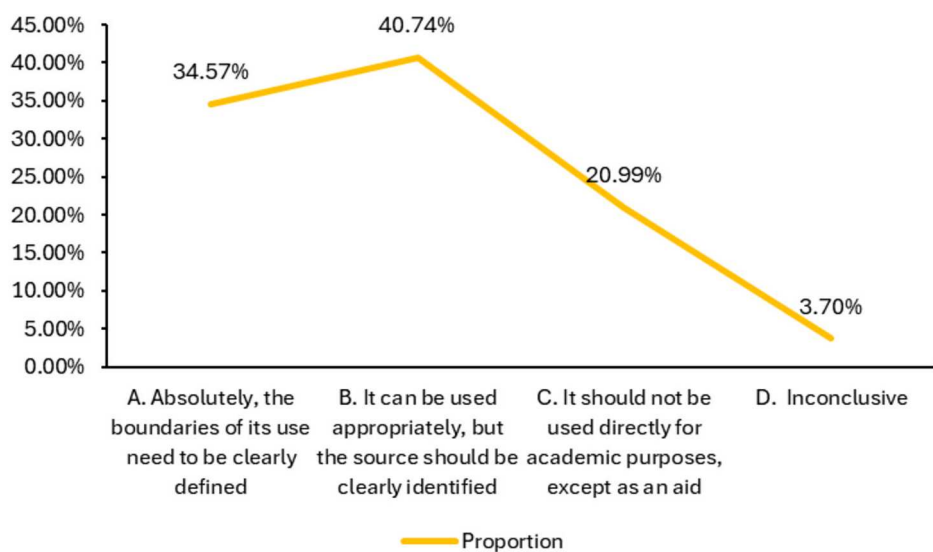


Figure 3. Statistics on whether the use of AI should comply with academic norms and ethical standards.

norms and ethical standards. In [Figure 3](#), 40.74% of respondents believe that AI-generated content can be used appropriately, provided that the source is clearly indicated. Meanwhile, 34.57% think that the boundaries of its use need to be clearly defined, while 20.99% hold the view that it should not be used directly for academic purposes, except as an aid. Only 3.70% of respondents are noncommittal.

[Figures 4–6](#) present various perspectives on whether content generated by artificial intelligence constitutes cheating. [Figure 4](#) indicates that 20.99% of respondents believe that directly using AI to generate content is cheating, while 49.38% feel that it depends on the specific circumstances. Additionally, 19.75% assert that acknowledging the source does not constitute cheating, and 9.88% find it difficult to define the issue.

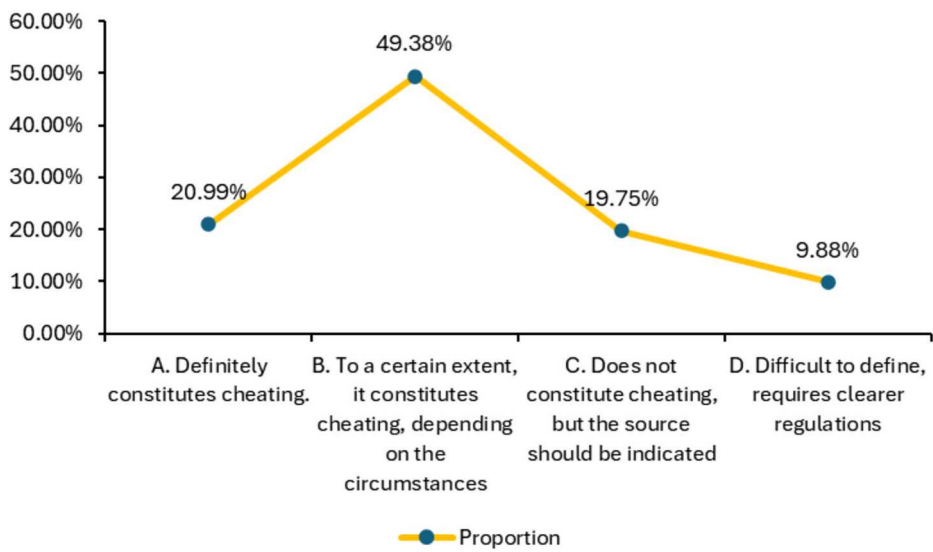


Figure 4. Statistics on whether the direct use of AI constitutes cheating.

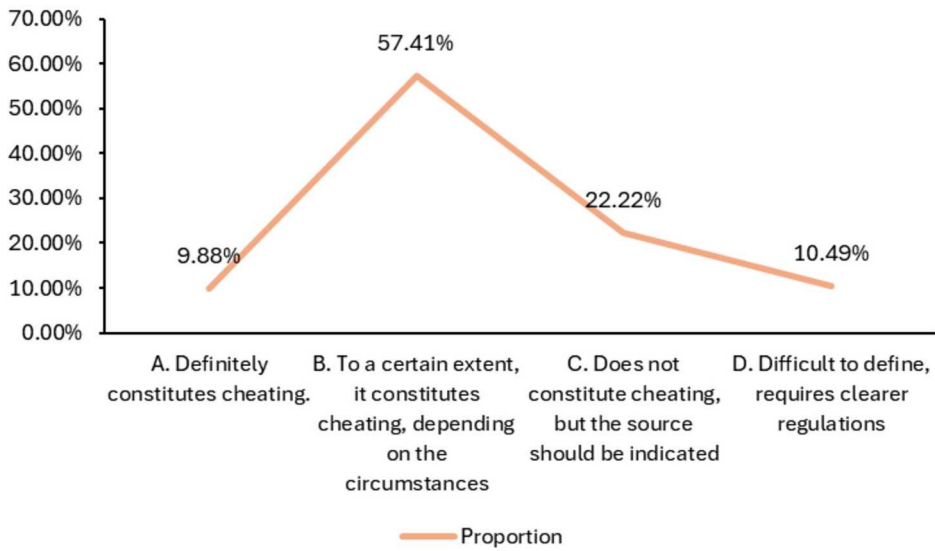


Figure 5. Statistics indicate whether minor modifications or manual adjustments to content generated by AI constitutes cheating.

Figure 5 examines the perceptions of manually modified AI-generated content. Among the respondents, 9.88% believe that such modifications constitute cheating, 57.41% think that depends on the specific circumstances, 22.22% assert that acknowledging the source does not equate to cheating, and 10.49% find it difficult to define the issue.

Figure 6 focuses on content that has been optimized exclusively using AI tools. Among the respondents, 2.47% believe that this practice definitely constitutes cheating, while 36.42% feel that it depends on the specific circumstances. Additionally, 48.15% of respondents assert that correctly citing sources does not constitute cheating, and 12.96% find it difficult to define the issue. As the level of AI participation decreases from “direct generation” to “auxiliary optimization,” the

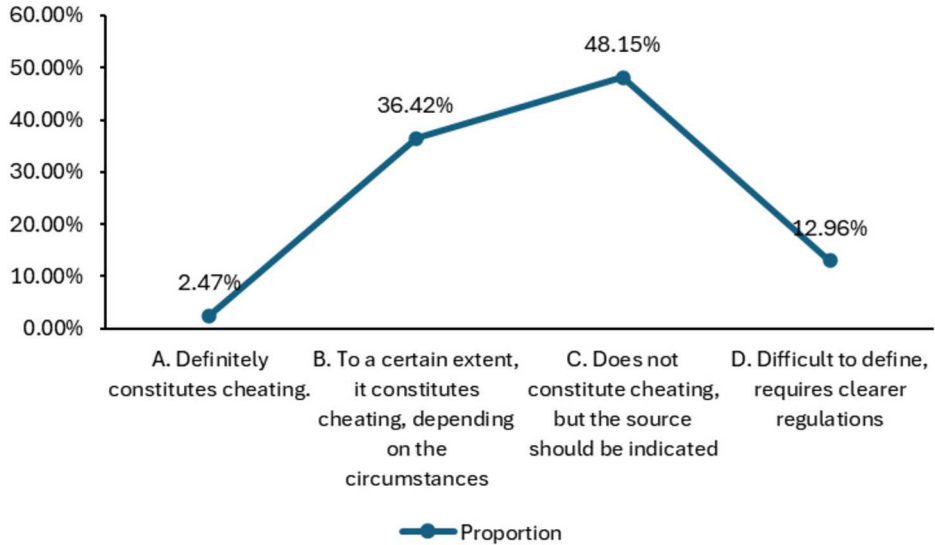


Figure 6. Statistics on whether the use of AI to polish and revise articles in an academic setting constitutes cheating.

percentage of individuals who perceive this as AI cheating has significantly declined (20.99% → 9.88% → 2.47%). Conversely, the proportion of those who believe that any AI involvement in annotation does not constitute cheating has notably increased (19.75% → 22.22% → 48.15%). This trend suggests that people have established moral standards to evaluate the extent of technological intervention (see Figures 3–6).

However, there are also concerns regarding AI-generated content. Critics worry that the rapid advancement of artificial intelligence will complicate the identification of infringements, prompting calls for regulations to be implemented as soon as possible. Some argue that if AI-generated content is used exclusively, it may undermine an individual's independent thinking and potentially diminish their cognitive abilities.

Taken together, the students' perspectives varied from ambiguous definitions of cheating to concerns about plagiarism and dishonesty. However, the majority of respondents still believe that the direct use of AI, or its involvement in significant modifications to article content, constitutes cheating. To ensure adherence to the principles of academic integrity and to clarify the definition of cheating, as well as to preserve individuals' capacity for critical thinking and comprehension, relevant guidelines must be developed in the future.

Is it necessary to develop specifications to guide AI-generated content?

According to the survey results, 93.21% of respondents believe it is necessary to develop relevant specifications to guide AI-generated content. 60.49% of respondents also believe that these specifications should be formulated carefully to avoid imposing excessive restrictions on innovation. Norms are established to maintain a positive academic environment, not to limit students' creativity. (See Figure 7) For instance, one respondent stated, "In my opinion, instructors should firmly instill in students the understanding that AI tools should only be used as supplementary resources." Students access information more quickly and efficiently, and even facilitate integration. However, the core ideas must always originate from humans – specifically, the students themselves – and not rely solely on AI tools. Artificial intelligence is developing rapidly, and as a new technology, its emergence presents challenges. The establishment of relevant norms should not only ensure the responsible use of artificial intelligence but also promote users' adaptability to these norms.

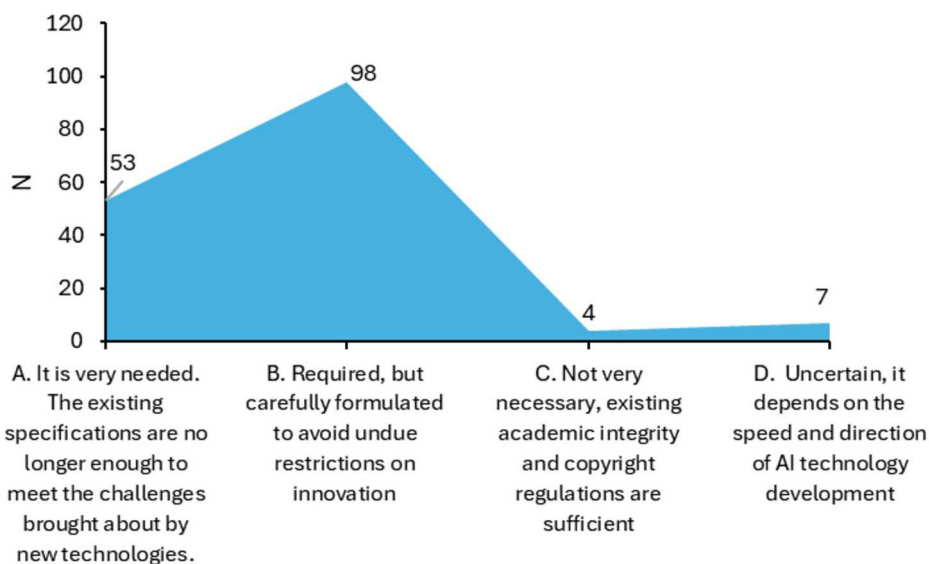


Figure 7. Views on whether there is a need for specifications to know AI-generated content.

If a school or teacher allows AI-assisted writing, how can integrity be ensured?

Based on the students’ reactions, most people believe that AI tools can sometimes be helpful in the creative process. Regarding the maintenance of academic integrity, it is essential to strengthen education on academic ethics while simultaneously establishing effective identification and detection mechanisms.

The survey results regarding measures to address the application of AI in academia, as illustrated in Figure 8, reveal that the most general option among respondents was “Clearly mark the AI-assisted creation part in the work,” that a significant majority considers clear identification to be an essential measure. 91 respondents selected “Strengthen academic ethics education and enhance self – restraint,” reflecting the importance people attach to raising awareness of academic ethics. 81 people chose “Establish a recognition and detection mechanism for AI – generated content,” suggesting that this is also a widely-endorsed approach. In contrast, only 50 respondents opted for “Mandated requirement for authors to manually review and edit content,” which is relatively fewer. And only 1 respondent chose “Other recommendations” (see Figure 8).

Students’ attitude towards the school’s maintenance of academic integrity

As shown in Figure 9, among all the options, 67 people, accounting for 41.36% of the respondents, chose “Support, but think implementation may be difficult.” 66 people, accounting for 40.74%, chose “Remain neutral and consider this just one of many norms.” These two results indicate that a relatively large proportion of respondents either support but are concerned about implementation difficulties or hold a neutral stance. The number of people choosing the other two options is relatively small, reflecting that respondents who clearly oppose or strongly support the school’s emphasis on the importance of academic integrity are both in the minority (see Figure 9).

Overall, the vast majority of respondents support schools emphasizing the importance of academic integrity. These surveys suggest that schools should actively collaborate with students to jointly safeguard academic integrity. However, the survey results also indicate that some students have varying levels of understanding regarding the significance of academic integrity education, and may require further promotion and guidance.

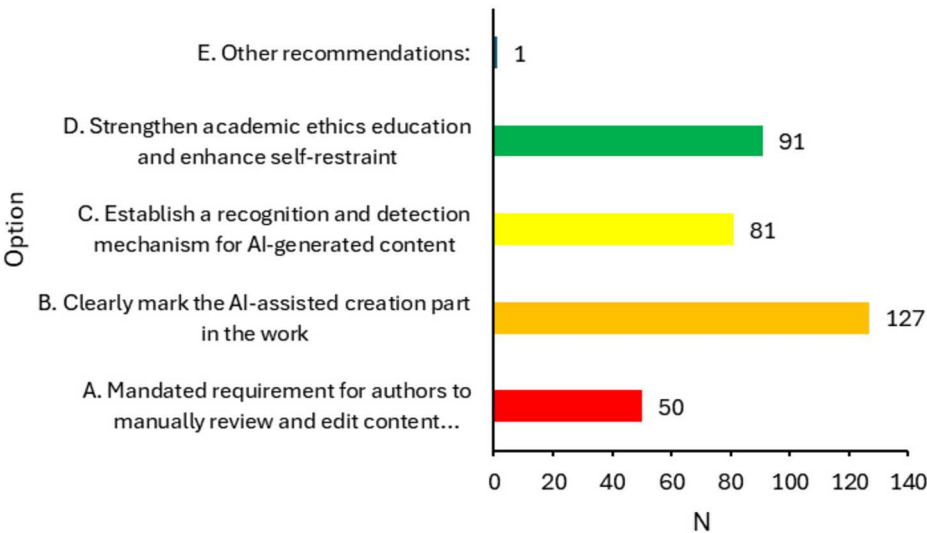


Figure 8. How to ensure academic integrity survey statistics.

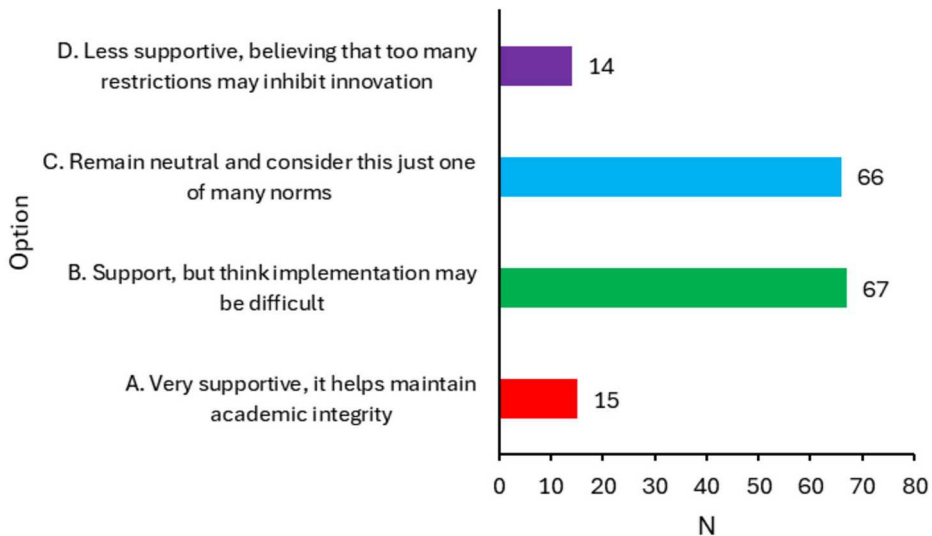


Figure 9. Statistics on students' perceptions of the school's maintenance of academic integrity.

Discussion

Artificial intelligence technology is developing rapidly and is regarded as a new era in the development of information technology, known as the fourth technological revolution. This technological revolution relies on the integration of various technological advances in information technology to simulate human cognitive activity and thinking ability. As a result, it gradually replaces and surpasses human mental work, offering humanity a path of liberation from the body to the brain (Van Noorden, 2022). However, it also brings new perspectives and challenges to humanity.

Pay attention to the subjective growth of teachers themselves

First of all, the current Chinese college students are Internet natives, and young people are easy to accept and learn new things. One of the biggest challenges for the teacher community is that teachers may not have as much knowledge about the application of AI technology as their students. In the future, academic integrity may be challenged with the advent of near-perfect AI tools. In response to this, teachers must determine how to identify student essays or assignments that violate academic integrity.

The use of ChatGPT in the education sector and its potential risks have received a lot of attention. Some schools explicitly prohibit students from using ChatGPT to answer exams and daily assignments to avoid cheating. Opinions are divided on whether ChatGPT and related generative AI technologies mean the future of education, with some promising for its ability to enhance learning and research, while others are skeptical that it could cause educators and students to become complacent and gradually abandon analytical skills (Cotton et al., 2024). Despite this, school bans have done little to discourage students from using generative AI in their academics and assignments (Futterer et al., 2023).

Traditionally, teachers have dominated teacher-student interactions. However, the rapid development of technology may challenge teachers' "pedagogical dignity" and subjectivity, leading to anxiety about self-identity and the recognition of others. However, it is also possible for teachers to maintain a constant motivation to learn, to be familiar with and master technological advances, and to achieve continuous development. In this context, the development of teachers' self-subjectivity is particularly important.

Teaching and demonstration of technology ethics

With the widespread use of generative AI technologies such as ChatGPT, students are exposed to a wealth of information. It is the teacher's responsibility to guide students to assess the authenticity of the information, extract the relevant information for problem-solving, and analyze the information logically and emotionally to draw conclusions (Jereb et al., 2018). This is a key area that teachers must prioritize. Teachers should not only set an example of technology ethics, but also guide students to understand the importance of technology ethics. This includes teaching subject matter knowledge and preventing the misuse of technological tools such as ChatGPT to shorten learning time. Teachers should also discourage inappropriate student behaviour when using technology.

As leading advocates, practitioners, and supervisors of the implementation of smart technologies, teachers' ethical considerations when using such technologies for teaching purposes have a direct impact on the efficiency and quality of teaching (Bretag, 2013). To ensure the responsible application of smart technology in education, teachers must adhere to the ethical guidelines for implementing smart technology. These guidelines fall under the umbrella of educational technology ethics, which combines the principles of educational ethics, pedagogical ethics, and technology ethics. Ethical standards evolve with technological advances and societal changes; Therefore, teachers must remain vigilant and adapt to emerging ethical dilemmas in their teaching methods (Šprajc et al., 2017).

ChatGPT provokes deep thought about fundamental questions in education.

"What kind of people to cultivate" is the most important concern in the field of education. According to Bloom's Classification of Learning Objectives in the Cognitive Domain (Gogus, 2012), memorization, comprehension, and application belong to lower-order thinking, which mainly involves the acquisition of specific knowledge, general principles, and concepts. Analyzing, synthesizing, and evaluating, on the other hand, are higher-level thinking activities that involve cognitive skills such as relational analysis, deduction, and judgment. When students are looking to get high scores through the existing testing system, they may unscrupulously use tools like ChatGPT. However, this process can lead to a neglect of academic integrity. Conversely, when their goal is to improve higher-order thinking skills, they often show self-discipline and effectively use AI tools for personal growth. Environmental factors also play a crucial role. If scores become the only criterion that matters, peer pressure may prompt more students to abuse ChatGPT.

Teachers need to adjust their assessment of students' knowledge and abilities in a timely manner. An effective way to do this is to combine an oral and essay exam to test whether the student really understands what is being said in the essay (in cases where ChatGPT is not suitable). Applied learning could also be improved, such as more workshops and a greater focus on practice tests and internship results. It is essential to educate students in information literacy, scientific integrity, and the acquisition of knowledge and skills through deep reading, critical thinking, and scientific writing. At the same time, the appropriate and effective use of ChatGPT can improve the creativity of teaching practices (Farrokhnia et al., 2024). When used correctly, ChatGPT can promote "AI literacy" as an integrated teaching goal across all subjects. For example, teachers can prompt students to analyze reports of historical events generated by ChatGPT and assess the credibility and accuracy of their sources. Teachers can use ChatGPT to develop students' critical thinking skills. In addition, teachers can also encourage students to carefully study the logical reasoning of the information generated by ChatGPT, explore the nuances of rhetoric, and use ChatGPT to improve their problem-solving and question-asking skills.

Perceptions of different countries on the participation of AI in academic creation

Different cultural perspectives may lead to varying views on the use of artificial intelligence (AI) in academic creation and ethics. A study was conducted to assess the perceptions, concerns, attitudes, and ethical considerations surrounding the use of ChatGPT among Generation Z college students in

higher education in Peru. An online survey was administered to 201 higher education students who had experience using ChatGPT for academic purposes. The findings suggest that while higher education students possess knowledge and positive attitudes towards ChatGPT, this does not guarantee its effective adoption and use. The ethical issues raised by reliance on ChatGPT must be addressed through responsible use programs in higher education (Acosta-Enriquez et al., 2024). In Nepal, concerns have been raised regarding how higher education students use, experience, and perceive AI in academia. The results indicate that the misuse of this technology in Nepal's higher education system may negatively impact students' creativity, writing habits, skills, and their ability to conduct independent research (Ghimire et al., 2024). A study examining the perceptions of Generation Z students in the Philippines regarding the forms and ethics of artificial intelligence (AI) and the Internet of Things (IoT) highlights that the ethical challenges associated with AI pose risks to data privacy and security (Jabar et al., 2024). To inform the revision of the University of Liverpool's Code of Conduct for Academic Integrity, a study surveyed students' perceptions of generative artificial intelligence (GAI) technology. The results indicated that students require clear policies on the use of GAI and that these technologies should not be banned by universities. However, it is essential to ensure that different groups of students have equal access to these technologies (Johnston et al., 2024). Additionally, a study was designed to investigate the determinants of acceptance and use of academic support provided by AI tools among students, focusing on their influence on student satisfaction and academic achievement in Pakistan and Malaysia. The findings reveal that the increasing use of these AI tools has enhanced student satisfaction and significantly impacted student outcomes in both countries. Furthermore, student engagement and personal innovation did not significantly affect the use of AI tools by students in either country (Dahri et al., 2024).

Despite the cultural differences across various countries, students in many different countries generally believe that AI can be effectively integrated into higher education and academia. However, if AI is not utilized responsibly, it may negatively impact the academic community and pose ethical challenges. In some nations, students argue that AI-assisted instruction can enhance academic performance, and they think that the potential drawbacks of AI on individual creativity to be acceptable. This discrepancy in perception may stem from differing views on the benefits of AI. Countries with a higher acceptance of AI tend to believe that its advantages outweigh the associated risks, leading them to take proactive measures to mitigate those risks in pursuit of greater benefits and efficiencies. Conversely, nations that view AI as a threat to morality, ethics, and individual creativity may argue that human ingenuity cannot be matched by AI, and that unrestricted AI usage could infringe upon academic and personal privacy. For instance, Elizabeth M. Wolkovich, an author, faced accusations of using ChatGPT to write a manuscript, which ultimately led to its rejection. In response, Wolkovich expressed her dissatisfaction with these allegations and defended her integrity, arguing that the mere existence of ChatGPT has undermined the academic process and highlighted the harm it has caused to her scholarly work (Wolkovich, 2024). In light of the challenges posed by AI to educational standards and academic integrity, institutions around the globe have developed countermeasures following the introduction of ChatGPT. For example, the University of Cambridge has stated that content generated by AI platforms, including ChatGPT, will be classified as academic misconduct. In contrast, the University of Oxford adopts a more open stance, viewing ChatGPT as a valuable tool for both educators and students.

The differences in the acceptance of AI may stem from variations in educational systems and cultural traditions across different countries. A cross-sectional study examining Malaysian medical students' attitudes and readiness towards artificial intelligence (AI) revealed that 87.36% of the students believe AI will play a significant role in the healthcare field. However, due to advancements in AI, 32.55% of the students are unlikely to pursue a career in radiology. The majority of students (71%) believe that AI education will benefit their careers, while 69.44% agree that all students should be taught using AI. Approximately 44.5% of students believe that they will possess the knowledge required for AI-related jobs after graduation (Tung & Dong, 2023). Countries that are more adept at utilizing artificial intelligence often believe that its integration into academia is more readily accepted.

Other factors influencing perceptions of artificial intelligence

One study showed that respondents' age, gender, occupation, religious beliefs, belief in conspiracy theories, and previous experience (familiarity) with AI can significantly influence respondents' perception of AI, both as an opportunity and as a threat. (Bozkurt & Gursoy, 2023) shows that AI is still seen as a double-edged sword in everyone's eyes. However, through our research and research, we can improve the relevant norms, so that more people can be familiar with, accept, and rationally use artificial intelligence.

Impact of the results of the study

According to the survey results, most college students said that artificial intelligence is deficient in language processing and originality in academic creation, and the use of artificial intelligence in academic creation is allowed, but it is limited to revision and polishing. The use of artificial intelligence may also inadvertently infringe on the achievements and privacy of others, so it is necessary to indicate the part where artificial intelligence is used. On the basis of the previous research, the results of this study will ensure that artificial intelligence can better enter the field of education in the future, and guide the direction for us to continuously revise the norms and coordinate the use of artificial intelligence in academic creation in the future. In the future, the use of artificial intelligence can not only promote academic innovation, but also ensure academic ethics and respect for the privacy and achievements of others. It is expected that in the future, the use of artificial intelligence will be as smooth as the coordinated and healthy operation of vehicles on the road through regulations and people's awareness of traffic obedience.

Limitations of the research results

This survey uses a questionnaire survey for Chinese college students, but there will be problems in the authenticity of the questionnaire information and the limited sample size of the survey in the survey process. In addition, China has a vast land and rich culture, and the level of regional economic development and education is inconsistent. Even in different regions, there may be factors that affect the accuracy of the survey results. By consulting foreign literature and cases, due to the different education methods, living habits, and scientific and technological development levels of various countries, there will also be the problem of sample limitations to a certain extent.

Expectations and recommendations

The development and application of artificial intelligence is the trend of the times, and we are ready to welcome it into our lives. Education is of unspoken importance to individuals, societies and nations. Therefore, it is of great significance to fully explore the direction in which AI is going in the field of education. From the perspective of college students, this study considers the possible impact of AI-generated content and academic norms on us, and expects that in the future, a complete normative system can be established to make the use of AI more rational while fully respecting the rights of each user. Here are some suggestions to set the stage for future developments.

At the legal level, apply regulations are developed to address academic integrity issues through legal means and impose severe penalties for violations.

At the academic level, it is recommended to improve the awareness of academic integrity in the whole society, and encourage original works by publicizing the concept of academic integrity and signing academic integrity agreements, so as to enhance the awareness of academic ethics.

At the level of education, it is appropriate to educate students on academic creditworthiness through educational means, strengthen education on academic creditworthiness, and when necessary, restrict or prohibit students from using artificial intelligence tools.

At the technical level, it is recommended to establish a corresponding screening system, including paper plagiarism checking, artificial intelligence real-name argumentation, etc., to minimize students' over-reliance on artificial intelligence tools in academic papers. It is advisable to question the basic content of the use of AI tools to determine if there is a situation of over-reliance.

At the level of AI development, a study based on social psychology showed that when confronted with generative AI's ability to reproduce the complexity of human cognitive abilities, participants reported significantly higher negative emotions than the control group. Conversely, the negative sentiment elicited by specific types of AI, such as generative AI, is associated with threat perception that extends to the AI technology as a whole (Alessandro et al., 2024). Therefore, the development of artificial intelligence can not be rushed, considering the social stability and changes in human psychology, we should gradually improve the acceptance of artificial intelligence changes at the same time, so that the changes of artificial intelligence can reasonably appear in people's lives.

In addition, the importance of establishing a system of rewards and punishments, such as the revocation of awards, the cancellation of relevant academic achievements, and the severe reprimand and punishment of violators of academic integrity, including delaying graduation and documenting misconduct, was also emphasized. This comprehensive system of advising helps uphold the fundamental principles of academic integrity at multiple levels.

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Data availability statement

All data generated or analyzed during this study are included in this published article.

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